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BAGONG PILIPINAS

CC 101 – Introduction to Computing

Laboratory Activity No. 1 | Week 2: Introduction to Computing (CO1) | Unit 2, Computing Career Pathways

Explore Career Pathways

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Course/Program: BSCS 1-1	**Date: September 16, 2026**

I. Objectives

At the end of this activity, you should be able to:

- Identify the major computing career pathways introduced in lecture and describe what each one involves.
- Choose one computing career pathway that best matches your interests and explain why.
- Map a realistic entry-to-expert roadmap for your chosen pathway, including representative job titles and key skills at each stage.
- Compare your current skills against your chosen pathway's requirements and identify a concrete plan to close the gap.

II. Materials Needed

- Your CC 101 Unit 2 lecture slides/notes on Computing Career Pathways
- This worksheet (printed copy, or the digital file to fill out and submit)
- Optional: internet access, for Part E (real job posting search)

III. Background

Computing is not just "programming." As introduced in the Unit 2 lecture, the field branches into several distinct career pathways, each with its own operational scope, required technical competencies, representative job titles, and professional standard or reference body: Software Engineering, Systems & Cloud Infrastructure, Data Science & AI Engineering, Cybersecurity & Threat Defense, and Information Systems & Enterprise IT. Each pathway also has its own career roadmap — the entry-level, mid-level, senior-level, and expert/leadership roles a professional typically grows through. This activity asks you to review those five pathways, choose the one that interests you most, and map out what growing in that pathway could look like for you.

IV. Procedure

Part A – Recall the Five Pathways

1. Using your lecture notes or slides, complete Table 1 by writing the core operational scope and one to two representative job titles for each of the five computing career pathways.

Part B – Choose Your Pathway

2. From the five pathways, choose ONE that interests you the most.
3. In two to three sentences, explain why you chose it (e.g., skills you already enjoy, a subject you like, a job title that excites you).

Part C – Map Your Career Roadmap

4. If your chosen pathway is Software Engineering, Systems & Cloud Infrastructure, Data Science

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& AI Engineering, or Cybersecurity & Threat Defense, use the career roadmap shown for it in the lecture. If you chose Information Systems & Enterprise IT (no roadmap was shown in lecture), research typical entry-level, mid-level, senior-level, and expert/leadership roles for this pathway online.

5. Complete Table 2 with example job titles and key skills/responsibilities at each of the four career stages.

Part D – Skill Gap Self-Assessment

6. Looking at the essential technical competencies for your chosen pathway (Table 1 / lecture), list at least five in Table 3. For each, mark whether you already have some experience with it, and briefly note how you plan to learn it if you don't.

Part E – Real-World Job Search (Optional / Bonus)

7. Search a real job platform (e.g., LinkedIn, Indeed, JobStreet, Kalibrr) for one entry-level job posting that matches your chosen pathway. Record it in Table 4.

V. Data and Findings

Table 1. The Five Computing Career Pathways (Recap — Part A)

Pathway	Core Operational Scope	Representative Job Title(s)
Software Engineering	Design, build, test, and maintain software apps and systems	Software Developer, Application Engineer
Systems & Cloud Infrastructure	Manage servers, networks, cloud platforms, and IT operations	Cloud Engineer, Network Administrator
Data Science & AI Engineering	Engineering Analyze data, build AI models, and create intelligent systems	Data Analyst, AI Developer
Cybersecurity & Threat Defense	Protect systems, networks, and data from attacks; prevent & respond to threats	Cybersecurity Analyst, Security Specialist
Information Systems & Enterprise IT	Manage business IT, align tech with organizational goals, support operations	IT Manager, Systems Analyst

My Chosen Pathway (Part B)

I am most interested in:

Why I chose it: I

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I want to join the Armed Forces of the Philippines and serve our country through cybersecurity. Protecting our nation's digital systems from threats is a meaningful and honorable way to contribute. I enjoy solving problems and protecting people, and this pathway lets me do both while defending our homeland. It is a career that matters deeply to me and gives me a strong sense of purpose.

Table 2. My Career Roadmap for My Chosen Pathway (Part C)

Career Stage	Example Job Title(s)	Key Skills / Responsibilities
Entry-Level	Cybersecurity Junior Analyst, IT Security Support	Monitor networks for unusual activity; apply basic security updates; follow safety protocols; help with security reports
Mid-Level	Cybersecurity Analyst, Security Operations Specialist	Detect and respond to threats; run security scans; manage firewalls; investigate breaches; create security plans
Senior-Level	Senior Security Analyst, Cybersecurity Engineer	Design strong security systems; lead threat response teams; research new attack methods; train junior staff; manage security projects
Expert / Leadership	Chief Information Security Officer, Cybersecurity Director	Set national or organizational security strategy; make key security decisions; lead large defense teams; protect critical infrastructure

Table 3. Skill Gap Self-Assessment (Part D)

Essential Competency	Already Have It? (Y/N)	How I Plan to Learn It
Basic computer & network knowledge	Y	Keep practicing and learning more advanced concepts
Understanding cyber threats & attacks Study cybersecurity lessons, watch tutorials, read news on digital safety	N	Study cybersecurity lessons, watch tutorials, read news on digital safety
Firewall & security tool operation	N	Follow hands-on exercises; use free practice tools online
Ethical hacking & defense basics	N	Take certified beginner courses; learn through school activities

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Essential Competency	Already Have It? (Y/N)	How I Plan to Learn It
Clear & accurate security reporting	N	Practice writing simple reports; improve my communication skills in class

Table 4. Real Job Posting Found (Part E, Optional / Bonus)

Item	Details
Job Title:	Cybersecurity Analyst / Information Security Officer
Company	Armed Forces of the Philippines — Cyber Defense Unit
Platform / Source (e.g., LinkedIn, Indeed, JobStreet)	AFP Careers Portal / Government Job Board
Required Skills Listed (3–5)	1. Network security knowledge 2. Threat detection & response 3. Basic digital forensics 4. Strong discipline & integrity 5. Teamwork & communication

VI. Realization (Reflection)

In two to four sentences each, answer the following in your own words.

1. Why did you choose this particular pathway over the other four?

I chose cybersecurity because it directly connects to my dream of serving in the AFP and protecting our country. The other pathways are important, but this one matches my goal of defending our nation's digital space. It gives me a clear and meaningful purpose — to keep our people, government, and institutions safe from harm.

2. Looking at your roadmap, which career stage excites you the most, and why?

The expert/leadership stage excites me the most because that is when I can lead and protect our country at the highest level.

3. Which required skill or competency for this pathway do you already have, and which one will be hardest for you to develop?

I already have basic computer knowledge and a strong willingness to learn and do what is right. The hardest skill to develop will be advanced threat detection and technical defense strategies, since they require deep technical knowledge and experience. But I am ready to study hard, practice often, and never give up until I master them.

4. How does mapping this roadmap change — or confirm — your plans for the rest of this course and your degree?

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It confirms that I am on the right path and gives me a clear direction to follow. Now I know exactly what skills I need to build step by step to reach my goal. It motivates me to take every lesson seriously and do my best in every activity. This roadmap reminds me that everything I am learning now is preparing me to serve and protect our country.

VII. Scoring Guide

Criteria	Description	Points
Completeness	Tables 1–3 and the reflection are fully filled out (Table 4 is bonus).	25
Roadmap Accuracy & Depth	Roadmap and skills recorded match the lecture pathway (or credible outside research for IS & Enterprise IT).	25
Realization / Reflection	Answers are thoughtful, specific to the student's own choice, and well written.	30
Neatness & Timeliness	Worksheet is legible/well-formatted and submitted on time.	20
Total		**100**

Checked by: **Score:** / 100

Instructor's Signature over Printed Name / Date

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